

OPERA NUMERORUM

NS Tower -- Path A: Phase 104
NS_Carleman_SmoothApprox_PROVED -- Friedrichs Mollification

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SECTION 1: THEOREM

SMOOTH DIV-FREE APPROXIMATION THEOREM (0 sorry, classical trio): Given: v in $L^2(\mathbb{R}^3; \mathbb{R}^3)$ with $\text{div}(v) = 0$ distributionally. Let ϕ in $C^\infty_c(\mathbb{R}^3)$, $\phi \geq 0$, $\int \phi = 1$. Define: $\phi_\epsilon(x) = \epsilon^{-3} * \phi(x/\epsilon)$, $v_\epsilon(x) = (\phi_\epsilon * v)(x)$. CONCLUSION: (1) v_ϵ in $C^\infty(\mathbb{R}^3; \mathbb{R}^3)$ [smoothness] (2) $\text{div}(v_\epsilon) = 0$ [div-free preserved] (3) $\|v_\epsilon - v\|_{L^2} \rightarrow 0$ as $\epsilon \rightarrow 0$ [L^2 convergence]

SECTION 2: PROOF OF THREE CLAIMS

Claim	Mathematical Argument	Lean 4 API
(1) Smoothness	$D^\alpha v_\epsilon = (D^\alpha \phi_\epsilon) * v$. ϕ_ϵ in C^∞_c , v in L^2 (locally integrable). Convolution of C^∞_c with L^2 is C^∞ . ALL multi-index derivatives exist and are continuous.	<code>HasCompactSupport</code> <code>.contDiff_convolution_left</code> <code>ContDiff</code> ■ $(\phi * v)$
(2) Div-free	On WHOLE SPACE $\mathbb{R}^3$ (no boundary): $d/dx_i (\phi_\epsilon * v_i)(x)$ = $\int d/dx_i \phi_\epsilon(x-y) v_i(y) dy$ = $-\int d/dy_i \phi_\epsilon(x-y) v_i(y) dy$ = $\int \phi_\epsilon(x-y) d/dy_i v_i(y) dy$ [IBP] = $(\phi_\epsilon * \text{div}_i v)(x)$. Sum: $\text{div}(v_\epsilon) = \phi_\epsilon * \text{div}(v) = \phi_\epsilon * 0 = 0$.	<code>fderiv_convolution</code> + integration by parts (whole space: no boundary term)
(3) L^2 conv.	ϕ_ϵ is an approximate identity in $L^2(\mathbb{R}^3)$: For any v in L^p ($1 \leq p < \infty$): $\ \phi_\epsilon * v - v\ _{L^p} \rightarrow 0$ as $\epsilon \rightarrow 0$. Classical theorem: Friedrichs mollification in L^p . Uses: translation continuity in L^2 + dominated conv.	<code>tendsto_conv_left</code> (L^2 approx identity) <code>MeasureTheory.Lp</code>

SECTION 3: KEY INSIGHT -- WHY WHOLE SPACE MATTERS

The div-free preservation (Claim 2) is SPECIFIC to the whole space $\mathbb{R}^3$. Integration by parts produces NO boundary term because ϕ_ϵ has compact support. This is why the ESS argument works on $\mathbb{R}^3$ (Clay problem) rather than on a domain with boundary (where additional boundary conditions on the mollified field would be needed). On a bounded domain, the Leray-Hopf projector must be applied explicitly to restore div-free — on $\mathbb{R}^3$, Friedrichs mollification alone suffices.

SECTION 4: GAP TABLE -- Phase 104, 4 deps (5 -> 4)

#	Named Open Def	Status
1	NS_BlowupConcentration_OPEN	OPEN (NEXT) -- Aubin-Lions + $L^{\{3,\infty\}}$ compactness
2	NS_Carleman_LimitPass_OPEN	OPEN -- limit passage in Carleman argument
3	NS_CarlemanHeat_OPEN	OPEN CRITICAL -- Hormander pseudo-convexity for heat
4	NS_CarlemanDriftAbsorption_OPEN	OPEN -- blocked on heat (dep 3)

SECTION 5: INTEGRITY

Cumulative: 95(7)->98(10)->99(8)->100(8)->101(7)->102(6)->103(5)->104(4) Repository:
DavidFox998/navier-stokes New file: Towers/NS/NSPhase104SmoothApprox.lean CMI rules: 0 sorry, 0
axiom keyword, classical trio only M7 manifest: FROZEN (5b80b84d) #print axioms
NS_M6_CLOSED_v104 -> {propext, Classical.choice, Quot.sound}